

Why AI Destroys Critical Thinking

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Opinion Brief

Before AI or any technology even existed, students had to get information through reading books. They had to go through the tedious process of going to the library, finding the book that they needed, and looking for information that could help them with their assignments. As the years progressed, students were able to seek aid through a simple search on Google. Today, students are able to type one single prompt into Chat GPT and generate a detailed response.

However, students have figured out that they can use AI to complete assignments for them, cheat on tests, and so much more. They are able to generate an entire paper, solve a complex math problem, or write summarized notes from a textbook without having to put in any effort.

Therefore, AI destroys critical thinking because it eliminates necessary cognitive friction, replaces writing as thinking, creates an illusion of flawlessness, bypasses significant reading, and limits creativity.

First of all, AI limits critical thinking because it eliminates cognitive friction. According to psychologists, “Cognitive friction is the mental effort required to process information, make decisions, or complete tasks when interacting with products, services, or systems. It represents the resistance or difficulty users experience during these interactions. Various factors contribute to cognitive friction, including complex user interfaces, confusing navigation, unclear instructions, and conflicting information” (Claritee 1). In other words, it means your brain is in active learning.

Many examples of using cognitive friction include trying three different formulas until one of them works for a math problem, trying to figure out how to transition from the first body paragraph to the second body paragraph in an essay, or trying to understand a complex ten-page text by rereading the text and looking up words. All of these examples include a lot of time and effort, but they strengthen the brain.

On the other hand, using AI requires no cognitive friction whatsoever. Students often prefer this strategy of eliminating the friction because they want to make their life easier. While this does save time and energy, it is preventing the brain from getting stronger. This is because the brain becomes accustomed to instant gratification and loses its stamina for hard mental tasks. Therefore, when the struggle is eliminated, the learning is also eliminated.

Second of all, AI eliminates critical thinking because it replaces writing as thinking. The process of writing is a process of deep thinking made visible. When a blank document is open, the brain is filled with half-remembered facts, unformed impressions, and vague instincts. The brain does not automatically think of a polished essay and writes it down, it creates a rough draft to figure out how it wants it to be formatted.

Psychologists call this externalized cognition. By forcing a vague feeling into a concrete sentence, you subject it to a harsh reality check. You look at your own words on the page and make changes as you continue writing. This uncomfortable friction is the exact moment your brain rejects a weak argument and creates a stronger one.

Additionally, you are looking at a screen that tells you what you think about any topic. Because the text is persuasive, your brain adopts it through a psychological, called automation bias. You start believing that the AI's output is your view on the topic. In reality, you didn't form that perspective. You didn't learn how to think, but you just learned how to agree with a machine.

Next, AI eliminates critical thinking because it creates an illusion of flawlessness. Students are vulnerable to automation bias because they trust machine-generated output over human intuition. AI models write with absolute confidence and flawless grammar. This polished presentation tricks students into lowering their skepticism. They unknowingly turn in incorrect data and fake historical facts simply because the machine was polished, weakening their baseline fact-checking habits.

This creates a dangerous feedback loop of unchecked facts. Because verification requires manual effort, students treat the first AI output as the truth. Their baseline skepticism and media literacy does not show up. Instead of learning how to second guess claims, spot logical mistakes, or demand evidence, they blindly copy and paste misinformation into their assignments.

Therefore, this intellectual dependency shifts students from active thinkers to mere data couriers. When education rewards the final output rather than the messy process of discovery, students learn to prioritize the appearance of intelligence over actual understanding.

Likewise, AI limits critical thinking because students bypass significant reading through generated summaries. For example, they are faced with a twenty-page reading assignment, so they decide to plug the text into AI and ask for a three-bullet-point summary. They bypass the physical act of turning pages, tracking arguments, and processing dense academic points on their own.

By doing this, students miss out on tracking complex themes, noticing subtle subtext, and building long attention spans. They become dependent on short, oversimplified excerpts of information. This shortcut robs them of the opportunity to build stamina for complex texts and develop low synthesis skills.

When AI processes a book or an article, it strips out the intellectual foundation, the author's tone, rhetorical strategies, and structural transitions. Students are left with sterile data points detached from the context. This constant exposure to low effort summaries changes the human brain's reading circuit. It conditions students to skim rather than scan, turning deep readers into people who can no longer navigate ambiguity, follow evidence, or tolerate complex sentence structures.

Finally, AI destroys critical thinking because it limits creativity. When faced with a creative task, such as brainstorming a story idea, designing a project, or writing a hook, students type their prompt into AI and accept the first three options it generates. They use the machine as a replacement for human imagination.

Human creativity is born from unique personal experiences, weird associations, emotional depth, and even accidental intellectual mistakes. AI does not create, but it predicts the most likely word combinations based on past internet data. By relying on AI for inspiration, students bypass the vital process of brainstorming and divergent thinking.

This creates an environment of total artistic and intellectual conformity. When an entire classroom uses the same underlying technology to spark ideas, the resulting student work begins to look, sound, and feel identical. It flattens the creative playing field into a sterile average. Original student voices are filtered out in favor of predictable, standardized machine output.

The widespread adoption of AI in education threatens critical thinking by eliminating necessary cognitive friction, replacing writing as thinking, creating an illusion of flawlessness, bypassing significant reading, and limiting creativity. By prioritizing the convenience of AI-generated content over the mental struggle of learning, students risk becoming data consumers rather than independent thinkers.

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